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Naive Bayes for biomedical and food-safety text

Week 5: A strong, fast, wrong-but-useful model, on real recall notices.

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By the end of this week you should be able to

Learning outcomes served: LO2, LO3, LO5

- Derive the naive Bayes classifier from Bayes' rule and the conditional independence assumption.
- Explain precisely which assumption is "naive" and why the classifier works anyway.
- Apply Laplace smoothing and explain the zero-frequency problem it solves.
- Work in log space and explain the numerical problem that motivates it.
- Compare multinomial and Bernoulli event models and choose between them.

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The biological problem

Part 1 · Why anyone cares about this week's method

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Biological context

- The FDA publishes every food recall it classifies, with a free-text reason for the recall and a hazard classification: Class I where there is a reasonable probability of serious adverse health consequences, Class II where consequences are temporary or medically reversible, Class III where use is unlikely to cause adverse consequences.

- The data is public domain under CC0 and covers 2004 to the present. Predicting hazard class from the free-text reason is a genuine, useful, tractable text classification problem in food safety.

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The method

Part 2 · Concepts, then the mathematics

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The naive assumption

Features are assumed conditionally independent given the class. For text this is plainly false, since words co-occur. The classifier is nonetheless often accurate, because the argmax over classes can be right even when the probabilities themselves are badly wrong.

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Maximum a posteriori classification

Choose the class maximising P(class) times the product of P(feature|class). The denominator P(features) is the same for every class and can be ignored for ranking.

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Key concepts

Prior probability The class base rate estimated from the training data. On imbalanced recall data this matters a great deal.

Likelihood P(word|class), estimated by counting. Inspecting the highest-ratio words is the cheapest interpretability you will ever get.

Laplace smoothing Add one to every count. Without it, a single unseen word forces the whole posterior to zero regardless of all other evidence.

Log probabilities Multiplying hundreds of small probabilities underflows to zero in floating point. Summing logs is exactly equivalent and numerically stable.

Multinomial versus Bernoulli Multinomial counts word occurrences; Bernoulli records only presence or absence. Multinomial usually wins on longer documents.

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The naive Bayes classifier

$$\hat{c} = \operatorname{argmax}_c \; P(c) \cdot \prod_i P(x_i \mid c)$$

Choose the class maximising the prior times the product of per-feature likelihoods. The denominator P(x) is the same for every class and can be dropped for ranking.

$P(c)$ the class prior, estimated by counting

$P(x_i \mid c)$ the likelihood of feature i given class c

$\prod$ the product over all features, and the naive step: the features are assumed conditionally independent given c

argmax we need only the ranking, not calibrated probabilities

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Laplace smoothing

$$P(w \mid c) = \frac{\text{count}(w, c) + \alpha}{N_c + \alpha \cdot |V|}$$

Without it, one unseen word sets the whole product to zero and vetoes the classification regardless of all other evidence.

- α the smoothing strength; $\alpha = 1$ is Laplace, $\alpha < 1$ is Lidstone
- N_c total token count in class c
- $|V|$ vocabulary size
- $\alpha = 0$ reproduces the zero-frequency problem exactly

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Working in log space

$$\log P(c \mid x) \propto \log P(c) + \Sigma \log P(x_i \mid c)$$

Multiplying hundreds of probabilities below 0.1 underflows a double long before the end of a document. Summing logs is exactly equivalent and numerically stable.

- **∝** proportional to: the normalising constant is recovered with the log-sum-exp trick when a probability is needed
- **Σ** the sum replaces the product
- **why** a 300-word document multiplies 300 small numbers

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Multinomial and Bernoulli event models

Why a model with a false assumption is still worth teaching.

- Multinomial counts how many times each term occurs, and generally wins on longer documents
- Bernoulli records only presence or absence, and handles explicit non-occurrence of terms
- The naive independence assumption is plainly false for text, since words co-occur
- It classifies well anyway, because the argmax can be right even when the individual probability estimates are badly wrong
- The posteriors are therefore not calibrated, and are typically pushed towards 0 and 1. Treat them as a ranking
- Interpretability is real but shallow: the model has learned which words co-occur with which decision, not what hazard is

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Worked example

Part 3 · Classifying one recall notice

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Classifying one recall notice

- Suppose the training data gives P(Class I) = 0.45 and P(Class II) = 0.55, and a new notice reads "undeclared milk allergen not listed on label".
- From the counts, P("allergen" | Class I) = 0.081 and P("allergen" | Class II) = 0.006; P("undeclared" | Class I) = 0.074 and P("undeclared" | Class II) = 0.009. The word "label" is roughly equally likely in both and contributes almost nothing.
- Working in logs, the Class I score is log(0.45) + log(0.081) + log(0.074) + … and the Class II score is log(0.55) + log(0.006) + log(0.009) + …
- Even though Class II has the higher prior, the two allergen-related terms are worth more than three orders of magnitude between them, and Class I wins decisively.
- That is also the interpretability story: you can point at the two words that decided the case. What you cannot do is claim the model understands allergen risk. It has learned which words co-occur with which classification decisions, which is a different and much shallower thing.

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Naive Bayes is transparent about which features moved the decision, and silent about whether that reflects anything real.

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Try this interactively

Naive Bayes Word Probability Explorer

Which words drive the classification, and what does smoothing change?

`weeks/week05.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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The foundation-model connection

Part 4 · The first honest comparison

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The first honest comparison

- Text classification is exactly where a pretrained language model should help, because a bag of words throws away order, negation and synonymy. "No undeclared allergens found" and "undeclared allergens found" are nearly identical to naive Bayes.
- A clinical or general-purpose language model handles this properly.
- But run the comparison before assuming the answer. With tens of thousands of labelled notices and a strong vocabulary signal, naive Bayes is competitive, trains in under a second, needs no GPU, and can be audited word by word.
- If a transformer buys two points of F1 for a thousand times the compute and no interpretability, that is a decision to make deliberately, not by default.

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Interpretation and practical

Part 5 · What the result means, and what you will do with it

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The classifier learns that the word "listeria" strongly predicts Class I. Is it learning about pathogen risk, or about how FDA staff write notices? Design a test that would distinguish the two, and say what you would expect to see in each case.

Think like a scientist. Bring your answer to the laboratory.

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Naive Bayes on food recall notices

This week's laboratory · Build a text classifier on real regulatory data and audit what it learned. · 2 hours

- Pull recall records from the openFDA food enforcement endpoint. Take reason_for_recall as the text and classification as the label. Report the class balance and note the date range you pulled.
- Vectorise with CountVectorizer and fit MultinomialNB. Report accuracy, macro-F1 and the majority-class baseline.
- Extract the twenty words with the highest log-probability ratio between Class I and Class II. Read them. Do they describe hazard, or do they describe FDA drafting conventions?
- Compare alpha=0 (no smoothing), alpha=1 (Laplace) and alpha=0.01. Explain what happens at alpha=0 and why.
- Compare MultinomialNB with BernoulliNB and with TfidfVectorizer. Report all four combinations in one table.
- Split by year instead of at random: train on records before a cut-off year and test on records after it. Compare with the random split and explain the difference in terms of Week 3.

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Self-check

- A word in a new document never appeared in the training data for class C. Without smoothing, what happens to P(C | document)?
- A) It is slightly reduced
- B) It becomes exactly zero regardless of all other evidence
- C) It is unchanged
- D) The word is ignored

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Self-check

- Why does naive Bayes often classify well even though its independence assumption is clearly false for text?
- A) The assumption is nearly true for long documents
- B) Classification needs only the correct argmax, and the ranking of classes can survive substantial errors in the individual probability estimates
- C) Smoothing corrects the dependence
- D) It does not classify well

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References and further reading

U.S. Food and Drug Administration. openFDA Food Enforcement API (Recall Enterprise System, 2004 to present).

Geron, A. Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow. O'Reilly Media. ISBN 9781492032649.

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Questions

Next week: continue to Week 6.

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